Curriculum vitae

PERSONAL INFORMATION

Konstantinos Bessas

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konstantinos-bessas.github.io

Date of birth 9 March 1994 | Nationality Italian

ACADEMIC POSITIONS

Mar 2023 – Present

Postdoctoral Researcher (Assegnista di Ricerca) in Mathematics

Università di Pavia, Pavia (Italy)

Mentor: Professor Maria Giovanna Mora

EDUCATION AND TRAINING

Nov 2019 – May 2023

PhD in Mathematics

ISCED 8

Università di Pisa, Pisa (Italy)

Supervisor: Professor Matteo Novaga

Thesis title: “Nonlocal Models in Calculus of Variations”

PhD thesis defended on 02/05/2023

Dec 2016 – Apr 2019

Master of Science in Mathematics

ISCED 7

Università degli Studi di Parma, Parma (Italy)

Thesis supervisors: Professor Jan Kristensen, Professor Massimiliano Morini

Thesis title: “Oscillations and concentrations in sequences of gradients”

Grade: 110/110 cum laude

Sep 2013 – Dec 2016

Bachelor of Science in Mathematics

ISCED 6

Università degli Studi di Parma, Parma (Italy)

Thesis supervisor: Professor Massimiliano Morini

Thesis title: “The Mumford-Shah functional in one space dimension”

Grade: 110/110 cum laude

Sep 2008 – Jun 2013

School-leaving certificate

ISCED 3

Liceo Scientifico Giacomo Ulivi, Parma (Italy)

Grade: 100/100 cum laude

RESEARCH INTERESTS

- Calculus of Variations
- Nonlocal Interaction Problems
- Isoperimetric Problems
- Geometric PDEs
- Image Processing

SCHOLARSHIPS

Nov 2019 – Jan 2023 PhD scholarship at the Department of Mathematics of the University of Pisa

2013 – 2016 Winner of an INdAM scholarship covering the studies towards the Bachelor’s degree (Borsa di studio INdAM per l’iscrizione a corsi di Laurea in matematica)

GRANTS

Mar 2024 – Feb 2027 Young researchers grant 2023, 2023-0873, awarded by Fondazione Cariplo (200.000 €). Single PI.

VISITS (FUNDED)

Long research stays:

May 2022 – Jun 2022 Department for Applied Mathematics (University of Freiburg)
(scientific collaboration with Professor Patrick Dondl)

May 2018 – Jun 2018 Mathematical Institute (University of Oxford)
(work under the supervision of Professor Jan Kristensen on my master thesis)

Visits for scientific collaborations:

(01 - 13) 06/2026 Department for Applied Mathematics (University of Freiburg)
(invited by Professor Patrick Dondl and Professor Sören Bartels)

(01 - 17) 07/2025 Department for Applied Mathematics (University of Freiburg)
(invited by Professor Patrick Dondl and Professor Sören Bartels)

(16 - 20) 06/2025 CEREMADE, CNRS (Université Paris-Dauphine)
(invited by Professor Antonin Chambolle)

(21 - 23) 05/2025 University of Modena and Reggio Emilia
(invited by Professor Serena Guarino Lo Bianco)

(12 - 16) 05/2025 CEREMADE, CNRS (Université Paris-Dauphine)
(invited by Professor Antonin Chambolle)

(07 - 09) 05/2025 University of Pisa
(invited by Professor Matteo Novaga)

(03 - 07) 04/2025 University of Florence
(invited by Professor Giorgio Saracco)

(03 - 12) 12/2024 CEREMADE, CNRS (Université Paris-Dauphine)
(invited by Professor Antonin Chambolle)

(30/09 - 04/10) 2024 CEREMADE, CNRS (Université Paris-Dauphine)
(invited by Professor Antonin Chambolle)

(27 - 29) 03/2023 University of Naples Federico II
(invited by Professor Roberta Schiattarella)

(14 - 15) 12/2021 University of Bologna
(invited by Professor Berardo Ruffini)

Other visits:

Aug 2014 INdAM scholarship holders meeting (Università di Perugia)

INVITED TALKS

09/06/2026 *Generalized BMO-type seminorms and vector-valued Sobolev functions*
Oberseminar
University of Freiburg

29/10/2025 *Regularity aspects of a class of nonlocal total variation-based image denoising models*
Analysis and Applied Mathematics (AAM) Seminar
Kennesaw State University, (online)

08/07/2025 *On Liquid Drops Minimizing Nonlocal Energies*
Oberseminar
University of Freiburg

23/05/2025 *Nonlocal Sobolev spaces and perimeters*
Mathematical Analysis Seminar
University of Modena and Reggio Emilia

04/04/2025 *Regularity aspects of a class of nonlocal total variation-based image denoising models*
Mathematical Analysis Seminars
University of Florence

28/11/2024 *Isotropic and anisotropic nonlocal variational models*
Workshop on Geometric properties and regularity in variational problems
University of Pavia

16/09/2024 *On the Shape of Small Liquid Drops Minimizing Nonlocal Energies*
Elliptic and parabolic problems: analytic and geometric methods
University of Naples Parthenope

24/07/2024 *On the Shape of Small Liquid Drops Minimizing Nonlocal Energies*
2nd AMS-UMI International Joint Meeting
Special Session A9: Functional Inequalities and PDEs
University of Palermo

29/05/2024 *On the Shape of Small Liquid Drops Minimizing Nonlocal Energies*
Shape Optimization, Geometric Inequalities and Related Topics – II Edition
University of Salento

23/05/2024 *On the Shape of Small Liquid Drops Minimizing Nonlocal Energies*
International Conference on Elliptic and Parabolic Problems: GAETA 2024
Minisymposium MS-10: Regularity theory for variational problems
Hotel Serapo, Gaeta (Latina)

21/05/2024 *On the Shape of Small Liquid Drops Minimizing Nonlocal Energies*
Lions-Magenes days 2024
University of Pavia

20/10/2023 *On the Shape of Small Liquid Drops Minimizing Nonlocal Energies*
SHSU Analysis Seminar
Sam Houston State University, (online)

04/07/2023 *Long-range interactions in configurations of liquid droplets at equilibrium*
Seminari MAP: Metodi per l'Analisi e la Probabilità
University of Pisa

03/04/2023 *Nonlocal total variation denoising model with weighted L^1 fidelity*
Seminari di Matematica Applicata
University of Pavia

27/03/2023 *Nonlocal total variation denoising model with weighted L^1 fidelity*
Seminario di Analisi
University of Naples Federico II

25/11/2022 *Fractional total variation denoising model with L^1 fidelity*
Weekend di lavoro su Calcolo delle Variazioni
Grand Hotel Tettuccio, Montecatini Terme (Pistoia)

02/06/2022 *Fractional total variation denoising model with L^1 fidelity*
Oberseminar
University of Freiburg

CONTRIBUTED TALKS

09/09/2026 *Generalized BMO-type seminorms and vector-valued Sobolev functions*
The 17th International Conference on Free Boundary Problems: Theory and Applications 2026
Humboldt-Universität, Berlin

09/07/2026 *Generalized BMO-type seminorms and vector-valued Sobolev functions*
The 15th AIMS Conference
National and Kapodistrian University of Athens

25/05/2026 *Generalized BMO-type seminorms and vector-valued Sobolev functions*
International Conference on Elliptic and Parabolic Problems: GAETA 2026
Hotel Serapo, Gaeta (Latina)

24/01/2024 *Dealing with a nonlocal world from a variational perspective*
Incontro Annuale dei Ricercatori in Matematica UNIPV
University of Pavia

19/09/2023 *Non-local BV functions and a denoising model with L^1 fidelity*
Shape Optimization, Isoperimetric and Functional Inequalities
Bellavista Relax Hotel, Levico Terme (Trento)

POSTERS

11/02/2026 *Regularity aspects of a class of nonlocal total variation-based image denoising models*
XXXV Convegno Nazionale di Calcolo delle Variazioni
Hotel Mediterraneo, Riccione

11/07/2023 *Fractional total variation denoising model with L^1 fidelity*
Hausdorff School: "Analysis of PDEs: Variational and Geometric Perspectives"
Hausdorff Center for Mathematics, Bonn

04/10/2022 *Fractional total variation denoising model with L^1 fidelity*
Nonlocality: Analysis, Numerics and Applications
Lorentz Center, Leiden

TEACHING

Sep 2021 – May 2022 Teaching assistant for the course “Analisi Matematica 1” (Calculus 1)
(Università di Pisa, Bachelor’s degree in Mathematics)

Sep 2020 – Jun 2021 Teaching assistant for the course “Analisi matematica 2” (Calculus 2)
(Università di Pisa, Bachelor’s degree in Mathematics)

ORGANIZATION

(14 - 17) 09/2026 Organizer of the workshop “Nonlocality in image denoising and other variational problems”
(Website: `https://nimvar2026.github.io/`)

(05 - 06) 02/2025 Co-organizer of the workshop “Incontro Annuale dei Ricercatori in Matematica UNIPV”
(in collaboration with Davide Duma, Monica Montardini and Michele Schiavina)

(24 - 25) 01/2024 Co-organizer of the workshop “Incontro Annuale dei Ricercatori in Matematica UNIPV”
(in collaboration with Andrea Bondesan, Carlo Marcati, Silvia Paparini and Michele Schiavina)

Nov 2021 – Jan 2023 Co-organizer of the seminar series “Seminari MAP: Metodi per l’Analisi e la Probabilità”
(in collaboration with Daniele Barbera, Luca Briani and Luciano Sciaraffia)

Mar 2021 – Nov 2021 Co-organizer of the seminar series “Seminari MAP: Metodi per l’Analisi e la Probabilità”
(in collaboration with Luca Briani and Vincenzo Scattaglia)

SCIENTIFIC PROJECTS

Mar 2024 - Feb 2027	Single PI of the Fondazione Cariplo Project *Nonlocality in image denoising and other variational problems* (Young researchers grant 2023, 2023-0873)
2026	Participation to the INdAM GNAMPA Project *Metodi non locali classici e distribuzionali per problemi variazionali*
2024	Participation to the INdAM GNAMPA Project *Proprietà geometriche e regolarità in problemi variazionali locali e non locali*
2023	Participation to the INdAM GNAMPA Project *Variational problems for non-local functionals and operators*
Mar 2023 - Aug 2023	Participation to the PRIN 2017 Project *Variational methods for stationary and evolution problems with singularities and interfaces*
2022	Participation to the INdAM GNAMPA Project *Nonlocal phenomena in local problems*
2020	Participation to the INdAM GNAMPA Project *Minimal clusters and minimal partitions*

CONFERENCES, WORKSHOPS, SCHOOLS

(14 - 17) 09/2026	Nonlocality in image denoising and other variational problems University of Pavia
(07 - 11) 09/2026	The 17th International Conference on Free Boundary Problems: Theory and Applications 2026 Humboldt-Universität, Berlin
(06 - 10) 07/2026	The 15th AIMS Conference National and Kapodistrian University of Athens
(25 - 29) 05/2026	International Conference on Elliptic and Parabolic Problems: GAETA 2026 Hotel Serapo, Gaeta (Latina)
(09 - 13) 02/2026	XXXV Convegno Nazionale di Calcolo delle Variazioni Hotel Mediterraneo, Riccione
(10 - 14) 02/2025	XXXIV Convegno Nazionale di Calcolo delle Variazioni Hotel Mediterraneo, Riccione
(28 - 29) 11/2024	Workshop on Geometric properties and regularity in variational problems University of Pavia
(16 - 17) 09/2024	Elliptic and parabolic problems: analytic and geometric methods University of Naples Parthenope
(23 - 26) 07/2024	2nd AMS-UMI International Joint Meeting University of Palermo
(24 - 28) 06/2024	Vito Volterra Meeting 2024 Department of Mathematics, Sapienza University, Rome
(27 - 29) 05/2024	Shape Optimization, Geometric Inequalities and Related Topics – II Edition University of Salento
(22 - 24) 05/2024	International Conference on Elliptic and Parabolic Problems: GAETA 2024 Hotel Serapo, Gaeta (Latina)
21/05/2024	Lions-Magenes days 2024 University of Pavia
(12 - 16) 02/2024	XXXIII Convegno Nazionale di Calcolo delle Variazioni Hotel Mediterraneo, Riccione
(24 - 25) 01/2024	Incontro Annuale dei Ricercatori in Matematica UNIPV University of Pavia
06/12/2023	Incontri di Analisi Matematica tra Firenze, Pisa e Siena University of Florence

21/11/2023	Boundary value problems and applications – The legacy of Enrico Magenes University of Pavia
26/10/2023	XI Giornata MI-PV Department of Mathematics, Politecnico di Milano
(18 - 22) 09/2023	Shape Optimization, Isoperimetric and Functional Inequalities Bellavista Relax Hotel, Levico Terme (Trento)
(17 - 21) 07/2023	Nonlinear PDEs: Recent Trends in the Analysis of Continuum Mechanics Hausdorff Center for Mathematics, Bonn
(10 - 14) 07/2023	Hausdorff School: "Analysis of PDEs: Variational and Geometric Perspectives" Hausdorff Center for Mathematics, Bonn
(12 - 16) 06/2023	Calculus of Variations & Geometric Measure Theory Aula Magna - Polo Fibonacci (University of Pisa)
(29/05 - 02/06) 2023	Geometric Measure Theory La Casa della Gioventù universitaria, Bressanone (Bolzano)
(25 - 27) 11/2022	Weekend di lavoro su Calcolo delle Variazioni Grand Hotel Tettuccio, Montecatini Terme (Pistoia)
(04 - 07) 10/2022	Nonlocality: Analysis, Numerics and Applications Lorentz Center, Leiden
(05 - 09) 09/2022	Anisotropic isoperimetric problems & related topics Best Western Globus Hotel, Rome
(20 - 24) 06/2022	Isoperimetric Problems Department of Mathematics (University of Pisa)
(23 - 25) 05/2022	Workshop on Nonlinear Bending Department for Applied Mathematics (University of Freiburg)
(09 - 13) 05/2022	XXXI Convegno Nazionale di Calcolo delle Variazioni Bellavista Relax Hotel, Levico Terme (Trento)
(21 - 25) 02/2022	Free Boundary Problems and related Evolution Equations ESI Vienna (Online)
(08 - 09) 11/2021	Ricordando Luciano Modica Department of Mathematics (University of Pisa)
(13 - 14) 10/2021	A Pisan workshop in Geometric Analysis Centro di Ricerca Matematica Ennio De Giorgi (Pisa)
(06 - 10) 09/2021	Variational methods and applications Centro di Ricerca Matematica Ennio De Giorgi (Pisa)
(21 - 25) 06/2021	Calculus of Variations and PDEs: recent developments and future directions FIM - Institute for Mathematical Research (Online)
(22 - 26) 02/2021	Winterschool on Analysis and Applied Mathematics University of Münster (Online)
(15 - 17) 12/2021	Online workshop in Geometric Analysis University of Pisa & Scuola Normale Superiore di Pisa (Online)
30/10/2020	Incontri di Analisi Matematica tra Firenze, Pisa e Siena University of Siena (Online)
(03 - 07) 02/2020	XXX Convegno Nazionale di Calcolo delle Variazioni Bellavista Relax Hotel, Levico Terme (Trento)
(25 - 28) 11/2019	Recent trends in Geometric analysis and applications Centro di Ricerca Matematica Ennio De Giorgi (Pisa)
22/11/2019	Incontri di Analisi Matematica tra Firenze, Pisa e Siena Department of Mathematics (University of Pisa)

(18 - 19) 06/2018 Women in PDEs
Mathematical Institute (University of Oxford)

PEER REVIEW ACTIVITY

Provided peer review for the journal *Calculus of Variations and Partial Differential Equations*.

POPULARIZATION

17/12/2025 Speaker at the second edition of the *Research Day*
University of Pavia

26/09/2025 Sharper Night - European Researchers' Night
Presentation of the stand *Matematica e Image denoising*

PERSONAL SKILLS

Mother tongue Italian

Other languages

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	B2	B2	B2	B2	B2
	First Certificate in English (FCE) B2				

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user
Common European Framework of Reference for Languages

PUBLISHED PAPERS

2027 [5] **K. Bessas**, S. Guarino Lo Bianco, and R. Schiattarella. **2027**. "Generalized BMO-type semi-norms and vector-valued Sobolev functions". *Nonlinear Analysis* 276. Available online 2026, p. 30. DOI: https://doi.org/10.1016/j.na.2026.114311.

2025 [4] **K. Bessas** and G. C. Brusca. **2025**. "A Note on Non-local Sobolev Spaces and Non-local Perimeters". *La Matematica* 4.4, pp. 829–860. DOI: 10.1007/s44007-025-00175-9.

[3] **K. Bessas** and G. Stefani. **2025**. "Non-local BV functions and a denoising model with L^1 fidelity". *Adv. Calc. Var.* 18.1, pp. 189–217. DOI: 10.1515/acv-2023-0082.

2023 [2] **K. Bessas**, M. Novaga, and F. Onoue. **2023**. "On the shape of small liquid drops minimizing nonlocal energies". *ESAIM Control Optim. Calc. Var.* 29.86, p. 26. DOI: 10.1051/cocv/2023080.

2022 [1] **K. Bessas**. **2022**. "Fractional total variation denoising model with L^1 fidelity". *Nonlinear Anal.* 222.112926, p. 20. DOI: 10.1016/j.na.2022.112926.

PHD THESIS

2023 [T1] **K. Bessas**. **2023**. "Nonlocal Models in Calculus of Variations". PhD thesis. Università di Pisa. URL: https://etd.adm.unipi.it/t/etd-04232023-211355/.

REFEREES

Antonin Chambolle chambolle@ceremade.dauphine.fr
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Dipartimento di Matematica, Università di Pavia

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Dipartimento di Matematica, Università di Pisa